$[\mathbf{a}^-, \mathbf{a}^+] = 1$.

We can make the Hamiltonian in Eq. 10.22 even simpler by defining a new operator,

$$\mathbf{N} = \mathbf{a}^+\mathbf{a}^-,$$

called the *number operator*. Once again, this is just a name, but as we'll see, it's a very good name. Stated in terms of the number operator, the Hamiltonian becomes

$$\mathbf{H} = \omega\hbar(\mathbf{N} + 1/2). \qquad (10.23)$$

So far, all we've done is define some symbols, $\mathbf{a}^+$, $\mathbf{a}^-$, and **N**, that make the Hamiltonian look deceptively simple; it's not clear that we are actually any closer to figuring out the energy eigenvalues. To proceed further, let's recall my earlier advice: whenever you see two operators, commute them. In this case, we already know one commutator:

$$[\mathbf{a}^-, \mathbf{a}^+] = 1. \qquad (10.24)$$

Next, let's find the commutator of the raising and lowering operators with the number operator **N**. We'll do this by brute force. Here are the steps:

$$[\mathbf{a}^-, \mathbf{N}] = \mathbf{a}^-\mathbf{N} - \mathbf{N}\mathbf{a}^- = \mathbf{a}^-\mathbf{a}^+\mathbf{a}^- - \mathbf{a}^+\mathbf{a}^-\mathbf{a}^-.$$

Now, we'll combine the terms in the form

$$[\mathbf{a}^-, \mathbf{N}] = (\mathbf{a}^-\mathbf{a}^+ - \mathbf{a}^+\mathbf{a}^-)\mathbf{a}^-.$$

This looks complicated until we notice that the expression in the parentheses is just $[\mathbf{a}^-, \mathbf{a}^+]$, which just happens to be 1. Using this fact to simplify, we get

$$[\mathbf{a}^-, \mathbf{N}] = \mathbf{a}^-.$$